

Strategy Decision Memo: Next Research Lane for Coinbase RTB Bot

Executive verdict

The repo's current state hard-limits what can be researched credibly next: it is optimized for **single-symbol spot paper operation**, with a **hard-coded breakout state machine**, **misaligned config vs runtime risk/slippage**, and **no research-grade backtest / walk-forward / sweep / multiple-testing subsystem**.
filecite\turn0file1

Given those constraints, the best next move is:

Primary lane: Lane A (OHLCV-only, spot-only, multi-asset rule-library / portfolio / ensemble research).
Backup lane: Lane D (BTC-only indicator mean reversion fallback) — strictly as a cheap baseline / negative-control style pretest, not as “the new strategy.” filecite\turn0file0

The ranking is about *current project state*, not theoretical “best edge”:

- 1) **Lane A** — most compatible with spot + candle data, and the only lane where the **research plan's statistical standards** can plausibly be met by *aggregating trade counts across many assets* while controlling multiple testing.
- 2) **Lane D** — easiest to falsify quickly, but BTC-only single-rule indicator hunting is structurally fragile and already has negative evidence from your own process. filecite\turn0file0
- 3) **Lane B** — mechanism-backed, but operationally blocked by eligibility/onboarding, margin/liquidation risk, and a likely architecture fork away from the current spot/paper scaffold. 1
- 4) **Lane C** — highest potential information content, but the engineering lift (L2 state, sequencing, storage, replay simulation) is disproportionate to current infrastructure and two-week decision timelines. 2

Final posture: **Build now**, but “build” means *build the research harness for Lane A*, not more live/paper runtime features and not more BTC-only pattern-chaining.

Capability-to-edge matrix

Repo facts that constrain strategy choice

Only the constraints that matter for lane choice:

- The current bot is effectively **single-symbol, spot-oriented**, and **paper-mode oriented**.
filecite\turn0file1
- The breakout logic is still a **hard-coded breakout-retest state machine** (not a modular strategy layer). filecite\turn0file1

- **Config vs runtime risk/slippage are misaligned** (risk sizing / slippage assumptions are not consistently config-driven). `filecite\turn0file1`
- **No research-grade evaluation subsystem exists** (no walk-forward engine, no parameter sweeps, no multiple-testing defense pipeline). `filecite\turn0file1`

Lane-by-lane matrix

Lane	Edge mechanism	Best supporting evidence	Strongest evidence against it	Exact required data	Exact required exchange / instrument access	F r
A: OHLCV-only, spot-only, multi-asset rules + portfolio	Cross-asset breadth + selection discipline + diversification: evaluate a large rule library, use formal multiple-testing control, then combine survivors into simple portfolios/ensembles to reduce single-model fragility. 3	Large-rule crypto studies report predictability/profitability after multiple-hypothesis procedures, with an explicit note that OOS differs by asset. 4	(1) Even with FDR, technical-rule “success” can be fragile: persistence fails and modest costs can wipe out gains in classic settings. 5 (2) BTC specifically shows mixed OOS evidence (incl. “no predictability for Bitcoin in OOS” in a comprehensive crypto rule study). 6	OHLCV candles per symbol: open/high/low/close/volume, plus timestamps. For Coinbase Advanced Trade candles: granularity includes <code>ONE_HOUR</code>, <code>FOUR_HOUR</code>, <code>ONE_DAY</code>; default/max 350 candles per request, so you must paginate. 7	Spot products only. Needs ability to list products and fetch candles via Advanced Trade REST. 8	H s d li r r s s h L s in r in □

Lane	Edge mechanism	Best supporting evidence	Strongest evidence against it	Exact required data	Exact required exchange / instrument access	F
B: derivatives carry / basis / funding	Structural returns from basis / carry (cash-and-carry: long spot, short futures; or funding capture). Edge comes from segmentation, constraints, and convenience yield dynamics—not indicator patterns. ¹¹	BIS documents large, time-varying crypto carry and ties it to demand for leveraged exposure and limits to arbitrage; it also reports that high carry predicts future crashes. ¹² Academic work on perpetuals formalizes funding mechanics and market structure. ¹³	The same BIS evidence says carry is linked to boom–bust dynamics and crash risk: “carry predicts crashes” means tail-risk is central, not incidental. ¹⁴ Real-world implementability is constrained by regulation, margin frictions, and liquidation mechanics (which are <i>the mechanism</i> of limits-to-arbitrage). ¹⁵	You need: (1) spot prices, (2) perp/futures prices, (3) funding rates (and their calculation timing), (4) margin/positions data for a realistic liquidation simulation, and (5) borrow/lend rates if doing cross-exchange or credit-based variants. ¹⁶	Requires derivatives trading capability. Coinbase Advanced Trade supports perpetual futures only in eligible regions , with onboarding and margin collateral requirements (USDC), plus leverage/margin health and liquidation thresholds. ¹⁵	F r

Lane	Edge mechanism	Best supporting evidence	Strongest evidence against it	Exact required data	Exact required exchange / instrument access	F
C: order-flow / order-book / microstructure	Alpha from information-rich microstructure state : order flow imbalance, book imbalance, short-horizon price impact, liquidity provision premia. ¹⁹	Coinbase-specific microstructure work finds order book state matters for price discovery and that limit orders contribute meaningfully because of their abundance. ²⁰ Cross-crypto evidence shows order flow can explain/predict returns (portfolio sorts, ML). ²¹	(1) Signal-to-noise is hard; LOB data is noisy and feature engineering matters. ²² (2) Engineering lift is large: must maintain a correct L2 book, handle sequencing, and build a replay simulator; otherwise you are fitting artifacts. Coinbase rate limits and channel semantics constrain scale. ²³	You need Level2 snapshots + updates (book depth), trade prints, and accurate timestamps; ideally store raw incremental updates to reconstruct state. Coinbase L2 channel provides snapshot/update with “new_quantity” updates and event_time; sequenced messages must be processed reliably. ²⁴	Requires live WebSocket consumption of <code>level2</code> (and likely <code>market_trades</code>) plus storage and replay; trading can be spot but research requires microsecond-to-second event handling. ²³	P re w o h a re □

Lane	Edge mechanism	Best supporting evidence	Strongest evidence against it	Exact required data	Exact required exchange / instrument access	
D: BTC-only indicator mean reversion fallback	Classical short-horizon reversal / mean reversion : fade extremes, volatility spikes, or intraday reversal regimes; purely OHLCV indicators, no order book. ²⁸	Academic work reports intraday return predictability that can include reversal components, conditional on events/liquidity/jumps. ²⁸	Your own documented process shows repeated BTC-only indicator-style strategies failing to meet even minimum viability gates, suggesting BTC at retail horizons may be structurally hard. Large-rule crypto evidence explicitly reports no OOS predictability for Bitcoin in at least one comprehensive setting. ²⁹	BTC-USD OHLCV at 5m–1h (depending on the exact mean reversion trigger), plus volume. For Coinbase, use candle endpoint granularities <code>FIVE_MINUTE</code> , <code>ONE_HOUR</code> , etc. ⁷	Spot-only is sufficient. If you want true “reversion” both ways (long/short), you’d need derivatives; but Lane D as written is spot-only, so you’re largely restricted to long-only or going to cash. ³⁰	

Primary lane and backup lane

Lane ranking for the current project state

Best → worst (in the current constraints of spot + paper + no research harness):

Lane A > Lane D > Lane B > Lane C.

This is not claiming Lane A is “most likely to work.” It is claiming Lane A is the only lane where (a) you can plausibly build a credibility path inside this repo’s constraints, and (b) the edge hypothesis can be falsified quickly without switching the entire product surface to derivatives or microstructure infrastructure.

Primary lane selection: Lane A

Lane A wins because it matches (1) what the repo can become with the least architectural discontinuity and (2) what the literature says is the *only defensible way* to pursue technical-rule alpha at all: **breadth + disciplined selection + multiple-testing control + OOS evaluation.** ³³

Mixed Bitcoin out-of-sample evidence (explicitly acknowledged)

The crypto technical-trading literature is not uniformly “bullish on BTC edge.” A comprehensive study of technical rules across cryptocurrencies reports *no* out-of-sample predictability for Bitcoin while finding predictability persists in other major cryptocurrencies. ²⁹ Separately, the large-rule-set Bitcoin-focused work by Deprez & Frömmel concludes that portfolios of selected rules can outperform buy-and-hold out-of-sample after costs, but also notes performance is stronger earlier in Bitcoin’s history and frames results as conditional on realistic costs and data-mining controls. ³⁴

So the honest read is: **mixed literature means mixed literature**. Lane A is chosen **not** because “BTC TA works,” but because (a) BTC-only single-rule iteration has already produced repeated failures in your own process, and (b) if technical-rule edge exists at all, the most plausible path is **cross-asset breadth** where selection and diversification reduce the single-asset fragility.

Why breadth + selection + multiple-testing control is non-negotiable

Testing many rules (or many parameterizations) on the same data creates **data snooping** risk: you will “discover” winners by chance unless you explicitly correct for it. White’s Reality Check is specifically framed as a method to test whether the best model found in a specification search truly has predictive superiority over a benchmark. ³⁵

Similarly, Bailey et al. argue that standard hold-out practices can be unreliable for investment backtests and propose a framework (CSCV) to estimate the **probability of backtest overfitting**. ³⁶

And importantly: even modern multiple testing can fail if you omit realistic trading costs; Anghel shows missing either transaction fees or liquidity costs can inflate false discoveries beyond acceptable levels. ³⁷

Those aren’t academic niceties; they are the difference between “research lane” and “strategy superstition.”

Why single-rule BTC backtests are structurally weak

Single-rule BTC-only backtests are structurally weak for three reasons:

- 1) **Non-stationarity / regime dependence**: what looks good in one regime frequently collapses in the next; this is consistent with the research plan’s repeated BTC strategy failures and its own interpretation that BTC may not respect indicator-defined boundaries reliably. ³⁸
- 2) **Low effective sample size**: one asset yields far fewer independent opportunities than a diversified universe; you get noisy inference and you can’t meet trade-count gating without bending the strategy into overtrading. ³⁹
- 3) **Specification search overfitting**: if you iterate filters until a curve looks good, PBO rises quickly and OOS decay becomes the default outcome. ³⁶

Lane A is not “more edge by default.” It is “the only lane where you can *measure* edge with enough breadth to make the research standards enforceable.”

Backup lane selection: Lane D

Lane D is the backup only because it is the **cheapest falsification** and a useful **baseline comparator** (a “if even this doesn’t work, your OHLCV-only BTC assumptions are in trouble” test). [filecite\turn0file0](#)

But it is not first because your own work already shows BTC-only indicator families failing, and at least one large-scale crypto technical study reports no Bitcoin OOS predictability.

Why the other two lanes are not first

Why Lane B is not first: It plausibly has a stronger mechanism narrative (carry/basis), but it is blocked by real operational gates: eligible-region onboarding, USDC margin collateral, margin health management and liquidation thresholds, and the need for a derivatives-aware execution/risk architecture. ³⁸ In your repo’s current paper/spot posture, Lane B is “new product surface,” not “new strategy.” [filecite\turn0file1](#)

Why Lane C is not first: Microstructure signals can be information-rich, but the repo does not yet have the engineering primitives: L2 book building, sequencing validation, a storage model for update streams, and a replay/backtest engine that simulates spread/queue/latency. Coinbase’s L2 feed exists and is well documented, but it is not a small incremental bridge from the current bar-based runtime. ³⁹

Why the previous search process was weak

The weakness was not “you didn’t try enough indicators.” It was that the process optimized for **building a bot** rather than **building falsifiable evidence**.

Key failure modes (each has a direct corrective in Lane A):

- **Strategy-first instead of capability-first:** the PDF explicitly notes the repo still hard-codes the breakout concept while the research plan says that concept is archived; that’s a process sign you kept “coding the next idea” inside runtime rather than isolating strategy research from orchestration.
- **BTC-only tunneling:** the research plan’s history shows repeated BTC-USD failures and an asset-level suspicion (“BTC-USD retail timeframes may be structurally difficult”)—yet the loop stayed BTC-only, single-rule-centric. [filecite\turn0file0](#)
- **Insufficient defenses against selection bias:** the research plan demands walk-forward + parameter sweeps + multiple-testing defenses, but the repo currently lacks that evaluation loop. This matters because data snooping is not hypothetical—White’s Reality Check was written specifically because repeated reuse of a single time series for model selection creates false positives by construction. ³⁵
- **Cost realism drift:** the PDF flags misalignment between config and runtime cost/risk assumptions; Anghel’s results imply that missing costs can directly bias false discovery control.
- **Conflating feasibility with evidence:** paper-mode orchestration being correct is not evidence of profitable edge; the PDF explicitly characterizes the repo as a paper/runtime harness, not a research platform. [filecite\turn0file1](#)

Lane A’s emphasis on *breadth* + *controlled selection* + *OOS gates* is the direct antidote.

Exact next 7 engineering tasks

These are ordered to maximize two-week falsifiability for Lane A (while keeping changes localized and avoiding a runtime rewrite).

- **Build a candle downloader + local cache (parquet/arrow) for multi-asset OHLCV** using the Advanced Trade “Get Product Candles” endpoint, including pagination handling (max 350 candles per request) and granularities needed for Lane A (4h/1d). ⁴⁰
- **Create a minimal research data model:** `Candle(symbol, ts_start, open, high, low, close, volume, granularity)` plus a dataset manifest (symbols, time coverage, missingness checks). This enables deterministic re-runs. `filecite\turn0file0`
- **Implement a bar-based backtest engine (event-driven on candles)** that can replay candles and call a strategy function; execution model must include fees + slippage and enforce “no lookahead.”
- **Unify fee/slippage/risk parameters into a single config surface for research runs** (do not leave hidden constants in code paths). This is necessary because parameter sweeps are meaningless if some parameters are unreachable from the sweep. `filecite\turn0file1`
- **Implement walk-forward runner** (rolling windows with no leakage) that outputs per-window train-selected params and OOS metrics. This is explicitly demanded by your standards.
- **Implement parameter sweep runner with strict experiment accounting:** run manifests (git SHA, config hash, universe, cost model), and results logged to SQLite/CSV for later audit and multiple testing.
- **Add at least one multiple-testing control module** suitable for rule libraries: start with FDR-based selection (practical), and optionally add a White-style reality-check bootstrap or PBO estimate as a higher bar once the pipeline works. ⁴¹

Exact next 7 research tasks

These are the minimum research moves that make Lane A a falsifiable hypothesis rather than “try rules until something works.”

- **Pre-register the search budget:** define rule families (e.g., MA/RSI/channel breakouts/volatility bands), parameter bounds, and the maximum number of configs you will test per family. This is your first defense against implicit p-hacking. ³
- **Define the asset universe and liquidity filters** (spot-only): decide which Coinbase spot pairs qualify (e.g., minimum history length, minimum volume proxy, exclude new/illiquid listings). Use a reproducible product snapshot. ⁸
- **Define the benchmark set:** buy-and-hold per asset, cash-only, and a simple time-series momentum baseline so you can tell whether “complex rules” beat trivial exposures. ⁴²
- **Define the transaction-cost model before looking at results:** maker/taker fees assumption (or conservative taker-only), plus slippage; treat cost sensitivity as part of the hypothesis (edge must survive plausible friction). ⁴³
- **Run a pilot study on a small subset** (e.g., 10 assets × daily candles × 3 rule families) to confirm trade counts, holding times, and that the walk-forward pipeline isn’t leaking. Reject anything with too few trades to evaluate. `filecite\turn0file0`
- **Run the full Lane A experiment:** evaluate library → apply multiple-testing control (FDR) → select survivors → build simple equal-weight ensembles and measure OOS stability. ⁴⁴

- **Write down “selection uncertainty” explicitly:** report how many rules were tried, how many passed after costs, how many survive OOS, and how sensitive results are to small parameter changes—this is the difference between edge and curve-fit. ³²

Two-week kill criteria

These are not “long-term performance thresholds.” These are **decision gates** to stop wasting time if the lane is failing.

Kill criteria for the primary lane

Kill **Lane A** within two weeks if all of the following are true after you complete the minimum viable research harness and run the first full experiment:

- **No strategy family produces any survivor after costs + FDR + walk-forward** that beats the benchmark on risk-adjusted terms (even modestly) across *multiple* windows (not a single lucky segment). ⁴⁵
- Any “winner” is explainable as **selection bias**: it disappears when you (a) widen/narrow parameters slightly, or (b) change the train/test cut, which is consistent with high PBO / backtest overfit logic. ³⁶
- Survivors exist only by implicitly relying on unrealistic friction assumptions (the Anghel problem: missing costs → inflated discoveries). ³⁷

If Lane A fails these gates, the correct response is **stop and reassess the entire OHLCV-only assumption**, not “try yet another indicator.” ^{filecite{turn0file0}}

Kill criteria for the backup lane

Kill **Lane D** within two weeks (or sooner) if:

- The BTC-only mean reversion rule cannot produce **positive OOS expectancy after costs** in a strict walk-forward setting, or it is hypersensitive to small parameter changes (a classic curve-fit pattern). ³²
- The rule’s performance is concentrated in one volatility regime / event cluster and does not generalize across splits (consistent with regime instability). ²⁸

Lane D is only a baseline—if it fails, it strengthens the case to avoid BTC-only OHLCV indicator work as a primary path. ^{filecite{turn0file0}}

Final recommendation: build now, pretest only, or stop and reassess

Build now — specifically, build the **Lane A research harness** (candle dataset + backtest + walk-forward + sweep + multiple-testing control) and run the first disciplined multi-asset experiment.

Do **not** “build now” in the sense of expanding live trading or adding more state-machine logic. The current repo is correctly described as a strong paper/runtime harness but not a research platform; lane choice requires research-grade evaluation first. `filecite[turn0file1]`

1 11 12 14 17 18 **Crypto carry**

<https://www.bis.org/publ/work1087.htm>

2 23 24 25 26 27 39 **Advanced Trade WebSocket Channels - Coinbase Developer Documentation**

https://docs.cdp.coinbase.com/coinbase-app/advanced-trade-apis/websocket/websocket-channels?utm_source=chatgpt.com

3 10 33 35 41 **A Reality Check for Data Snooping | The Econometric Society**

https://www.econometricsociety.org/publications/econometrica/2000/09/01/reality-check-data-snooping?utm_source=chatgpt.com

4 6 29 42 **Technical trading and cryptocurrencies | Annals of Operations Research | Springer Nature Link**

<https://link.springer.com/article/10.1007/s10479-019-03357-1>

5 44 **Technical trading revisited: False discoveries, persistence tests, and transaction costs - ScienceDirect**

https://www.sciencedirect.com/science/article/pii/S0304405X1200116X?utm_source=chatgpt.com

7 9 40 **Get Product Candles - Coinbase Developer Documentation**

https://docs.cdp.coinbase.com/api-reference/advanced-trade-api/rest-api/products/get-product-candles?utm_source=chatgpt.com

8 **Get Product - Coinbase Developer Documentation**

https://docs.cdp.coinbase.com/api-reference/advanced-trade-api/rest-api/products/get-product?utm_source=chatgpt.com

13 **(PDF) Fundamentals of Perpetual Futures**

https://www.researchgate.net/publication/366287738_Fundamentals_of_Perpetual_Futures?utm_source=chatgpt.com

15 30 38 **Advanced Trade Perpetual Futures - Coinbase Developer Documentation**

<https://docs.cdp.coinbase.com/coinbase-business/advanced-trade-apis/guides/perpetual>

16 **US Perpetual-Style Futures Funding Rate Mechanism | Coinbase Help**

<https://help.coinbase.com/derivatives/perpetual-style-futures/funding-rate>

19 20 **Price Discovery in Bitcoin: The Role of Limit Orders by Carol Alexander, Daniel F. Heck, Andreas Kaeck :: SSRN**

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4150979&utm_source=chatgpt.com

21 **Order flow and cryptocurrency returns - ScienceDirect**

https://www.sciencedirect.com/science/article/pii/S1386418126000029?utm_source=chatgpt.com

22 **(PDF) Exploring Microstructural Dynamics in Cryptocurrency Limit Order Books: Better Inputs Matter More Than Stacking Another Hidden Layer**

https://www.researchgate.net/publication/392513905_Exploring_Microstructural_Dynamics_in_Cryptocurrency_Limit_Order_Books_Better_Inputs_Matter_More_Than_Stacking_Another_Hidden_La

[392513905_Exploring_Microstructural_Dynamics_in_Cryptocurrency_Limit_Order_Books_Better_Inputs_Matter_More_Than_Stacking_Another_Hidden_La](https://www.researchgate.net/publication/392513905_Exploring_Microstructural_Dynamics_in_Cryptocurrency_Limit_Order_Books_Better_Inputs_Matter_More_Than_Stacking_Another_Hidden_La)
[utm_source=chatgpt.com](https://www.researchgate.net/publication/392513905_Exploring_Microstructural_Dynamics_in_Cryptocurrency_Limit_Order_Books_Better_Inputs_Matter_More_Than_Stacking_Another_Hidden_La)

28 **(PDF) Intraday return predictability in the cryptocurrency markets: Momentum, reversal, or both**

https://www.researchgate.net/publication/361202793_Intraday_return_predictability_in_the_cryptocurrency_markets_momentum_reversal_or_both?utm_source=chatgpt.com

[361202793_Intraday_return_predictability_in_the_cryptocurrency_markets_momentum_reversal_or_both?](https://www.researchgate.net/publication/361202793_Intraday_return_predictability_in_the_cryptocurrency_markets_momentum_reversal_or_both?utm_source=chatgpt.com)
[utm_source=chatgpt.com](https://www.researchgate.net/publication/361202793_Intraday_return_predictability_in_the_cryptocurrency_markets_momentum_reversal_or_both?utm_source=chatgpt.com)

31 37 43 45 No pain, no gain: You should always incorporate trading costs for a bias-free evaluation of trading rule overperformance - ScienceDirect

https://www.sciencedirect.com/science/article/pii/S0165176522001720?utm_source=chatgpt.com

32 36 The Probability of Backtest Overfitting by David H. Bailey, Jonathan Borwein, Marcos Lopez de Prado, Qiji Jim Zhu :: SSRN

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2326253&utm_source=chatgpt.com

34 Are simple technical trading rules profitable in bitcoin markets? - ScienceDirect

https://www.sciencedirect.com/science/article/abs/pii/S1059056024003010?utm_source=chatgpt.com